HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed **C++** security software components for **AI accelerator cards**, focusing on **TDISP-based trusted device assignment** and lifecycle management.
- Implemented **SR-IOV-based secure execution environments**, enabling isolation and attestation of **PCIe virtual functions** for enhanced platform security.
- Integrated **PCIe security protocols software**, adhering to specifications from **DMTF (SPDM, CMA, MCTP)** and **PCI SIG (CMA, DOE, IDE)**.
- Developed **secure boot and attestation mechanisms** within the **Trusted Execution Environment**, enforcing granular security policies across the architecture.
- Optimized **memory management and performance** within secure enclaves, ensuring security guarantees were preserved during high-throughput operations.
- Debugged complex software issues in embedded Linux systems using **JTAG**, analyzing **RAM dumps**, and applying secure debug policies to resolve critical bugs.
- Contributed to **threat modeling and risk assessments**, aligning security architecture with **TDISP, OCP, TCG, and Confidential Computing Consortium** standards.

SKILLS

Languages – C++, C, Python, SQL, Java

Embedded Systems – Device Driver Development, RTOS, SoC Architecture, ARM TrustZone, RISC-V

Security Concepts – Trusted Execution Environments (TEE), Secure Boot, Attestation, Cryptographic Primitives, TPM, Platform Security, Threat Modeling

Protocols & Standards – TDISP, SPDM, CMA, MCTP, DOE, IDE, DICE, PCI SIG, DMTF

Tools & Technologies – Linux, Git, Gerrit, JTAG, RAM Dumps, Compilers, Profilers, Emulators, Logic Analyzers, Jenkins, Docker

Cloud & Virtualization – SR-IOV, PCIe, Virtual Functions, Confidential Computing

PROJECTS

Secure Boot and Attestation for Edge AI Accelerators

- Engineered a **secure boot process** for next-generation AI accelerator cards, integrating hardware root-of-trust mechanisms.
- Developed **attestation services** to verify the integrity of firmware and hardware components before and during runtime, reducing unauthorized access attempts by an estimated **40%**.
- Implemented **cryptographic primitives** for secure communication channels between the host system and the AI accelerator.

TEE Implementation for PCIe Device Isolation

- Designed and implemented **SR-IOV** configurations to create isolated virtual functions for PCIe devices, enhancing multi-tenancy security.
- Developed software components to manage the lifecycle and **attestation** of these virtual functions within a **Trusted Execution Environment**.
- Integrated low-level driver code with the RTOS to manage secure memory regions and enforce access controls for isolated devices.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)